9 Morphological Image Processing

In form and feature, face and limb,
I grew so like my brother
That folks got taking me for him
And each for one another.

Henry Sambrooke Leigh, Carols of Cockayne, The Twins

Preview

The word *morphology* commonly denotes a branch of biology that deals with the form and structure of animals and plants. We use the same word here in the context of *mathematical morphology* as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons, and the convex hull. We are interested also in morphological techniques for pre- or postprocessing, such as morphological filtering, thinning, and pruning.

In the following sections we develop and illustrate several important concepts in mathematical morphology. Many of the ideas introduced here can be formulated in terms of n-dimensional Euclidean space, E^n. However, our interest initially is on binary images whose components are elements of Z^2 (see Section 2.4.2). We discuss extensions to gray-scale images in Section 9.6.

The material in this chapter begins a transition from a focus on purely image processing methods, whose input and output are images, to processes in which the inputs are images, but the outputs are attributes extracted from those images, in the sense defined in Section 1.1. Tools such as morphology and related concepts are a cornerstone of the mathematical foundation that is utilized for extracting "meaning" from an image. Other approaches are developed and applied in the remaining chapters of the book.

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9.1 Preliminaries

You will find it helpful to review Sections 2.4.2 and 2.6.4 before proceeding.

The language of mathematical morphology is set theory. As such, morphology offers a unified and powerful approach to numerous image processing problems. Sets in mathematical morphology represent objects in an image. For example, the set of all white pixels in a binary image is a complete morphological description of the image. In binary images, the sets in question are members of the 2-D integer space Z^2 (see Section 2.4.2), where each element of a set is a tuple (2-D vector) whose coordinates are the (x, y) coordinates of a white (or black, depending on convention) pixel in the image. Gray-scale digital images of the form discussed in the previous chapters can be represented as sets whose components are in Z^3. In this case, two components of each element of the set refer to the coordinates of a pixel, and the third corresponds to its discrete intensity value. Sets in higher dimensional spaces can contain other image attributes, such as color and time varying components.

The set reflection operation is analogous to the flipping (rotating) operation performed in spatial convolution (Section 3.4.2).

In addition to the basic set definitions in Section 2.6.4, the concepts of set reflection and translation are used extensively in morphology. The *reflection* of a set B, denoted $\hat{B}$, is defined as

$$\hat{B} = \{w | w = -b, \quad \text{for} \quad b \in B\} \tag{9.1-1}$$

If B is the set of pixels (2-D points) representing an object in an image, then $\hat{B}$ is simply the set of points in B whose (x, y) coordinates have been replaced by $(-x, -y)$. Figures 9.1(a) and (b) show a simple set and its reflection.†

a b c

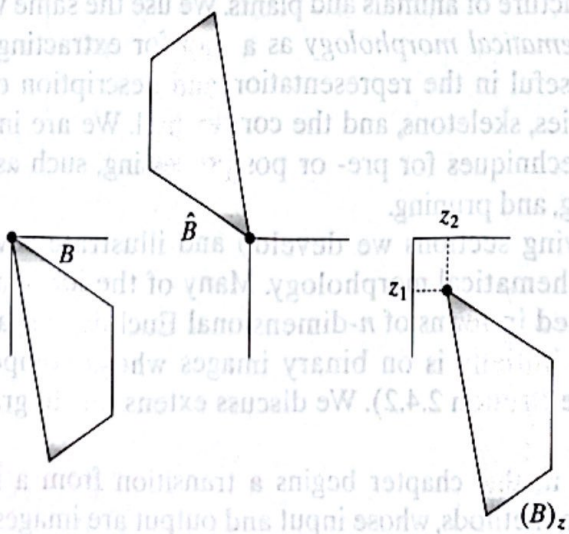


FIGURE 9.1 (a) A set, (b) its reflection, and (c) its translation by z.

†When working with graphics, such as the sets in Fig. 9.1, we use shading to indicate points (pixels) that are members of the set under consideration. When working with binary images, the sets of interest are pixels corresponding to objects. We show these in white, and all other pixels in black. The terms *foreground* and *background* are used often to denote the sets of pixels in an image defined to be objects and non-objects, respectively.

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The *translation* of a set B by point $z = (z_1, z_2)$, denoted $(B)_z$, is defined as

$$(B)_z = \{c | c = b + z, \quad \text{for} \quad b \in B\} \tag{9.1-2}$$

If B is the set of pixels representing an object in an image, then $(B)_z$ is the set of points in B whose (x, y) coordinates have been replaced by $(x + z_1, y + z_2)$. Figure 9.1(c) illustrates this concept using the set B from Fig. 9.1(a).

Set reflection and translation are employed extensively in morphology to formulate operations based on so-called *structuring elements* (SEs): small sets or subimages used to probe an image under study for properties of interest. The first row of Fig. 9.2 shows several examples of structuring elements where each shaded square denotes a member of the SE. When it does not matter whether a location in a given structuring element is or is not a member of the SE set, that location is marked with an "×" to denote a "don't care" condition, as defined later in Section 9.5.4. In addition to a definition of which elements are members of the SE, the origin of a structuring element also must be specified. The origins of the various SEs in Fig. 9.2 are indicated by a black dot (although placing the center of an SE at its center of gravity is common, the choice of origin is problem dependent in general). When the SE is symmetric and no dot is shown, the assumption is that the origin is at the center of symmetry.

When working with images, we require that structuring elements be rectangular arrays. This is accomplished by appending the smallest possible number of background elements (shown nonshaded in Fig. 9.2) necessary to form a rectangular array. The first and last SEs in the second row of Fig. 9.2 illustrate the procedure. The other SEs in that row already are in rectangular form.

As an introduction to how structuring elements are used in morphology, consider Fig. 9.3. Figures 9.3(a) and (b) show a simple set and a structuring element. As mentioned in the previous paragraph, a computer implementation requires that set A be converted also to a rectangular array by adding background elements. The background border is made large enough to accommodate the entire structuring element when its origin is on the border of the

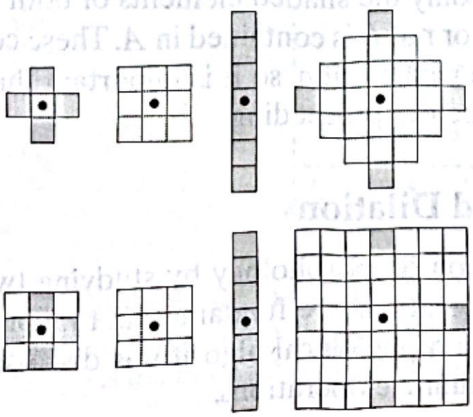
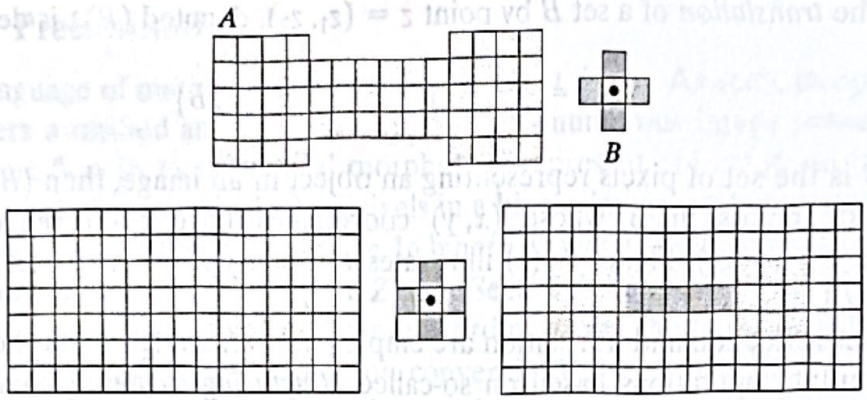


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

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a b
c d e

FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array and provide a background border. (d) Structuring element as a rectangular array. (e) Set processed by the structuring element.

In future illustrations, we add enough background points to form rectangular arrays, but let the padding be implicit when the meaning is clear in order to simplify the figures.

original set (this is analogous to padding for spatial correlation and convolution, as discussed in Section 3.4.2). In this case, the structuring element is of size 3×3 with the origin in the center, so a one-element border that encompasses the entire set is sufficient, as Fig. 9.3(c) shows. As in Fig. 9.2, the structuring element is filled with the smallest possible number of background elements necessary to make it into a rectangular array [Fig. 9.3(d)].

Suppose that we define an operation on set A using structuring element B, as follows: Create a new set by running B over A so that the origin of B visits every element of A. At each location of the origin of B, if B is completely contained in A, mark that location as a member of the new set (shown shaded); else mark it as not being a member of the new set (shown not shaded). Figure 9.3(e) shows the result of this operation. We see that, when the origin of B is on a border element of A, part of B ceases to be contained in A, thus eliminating the location on which B is centered as a possible member for the new set. The net result is that the boundary of the set is *eroded*, as Fig. 9.3(e) shows. When we use terminology such as "the structuring element is contained in the set," we mean specifically that the elements of A and B fully overlap. In other words, although we showed A and B as arrays containing both shaded and nonshaded elements, only the shaded elements of both sets are considered in determining whether or not B is contained in A. These concepts form the basis of the material in the next section, so it is important that you understand the ideas in Fig. 9.3 fully before proceeding.

9.2 Erosion and Dilation

We begin the discussion of morphology by studying two operations: *erosion* and *dilation*. These operations are fundamental to morphological processing. In fact, many of the morphological algorithms discussed in this chapter are based on these two primitive operations.

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9.2.1 Erosion

With A and B as sets in Z^2, the erosion of A by B, denoted $A \ominus B$, is defined as

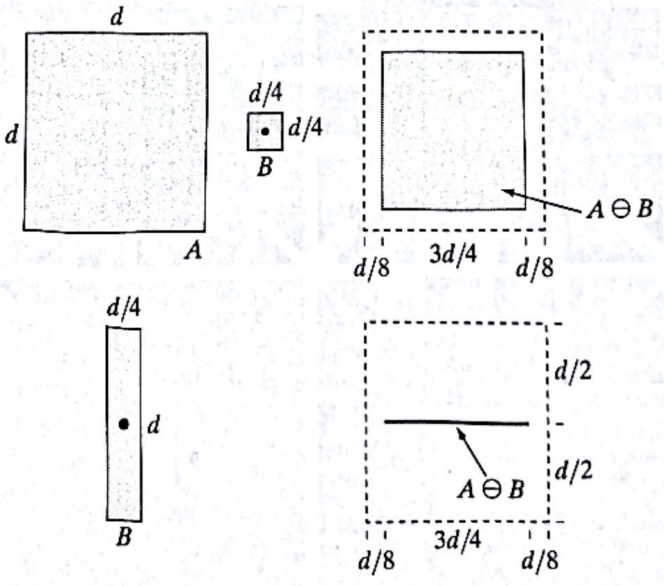
$$A \ominus B = \{z | (B)_z \subseteq A\} \tag{9.2-1}$$

In words, this equation indicates that the erosion of A by B is the set of all points z such that B, translated by z, is contained in A. In the following discussion, set B is assumed to be a structuring element. Equation (9.2-1) is the mathematical formulation of the example in Fig. 9.3(e), discussed at the end of the last section. Because the statement that B has to be contained in A is equivalent to B not sharing any common elements with the background, we can express erosion in the following equivalent form:

$$A \ominus B = \{z | (B)_z \cap A^c = \varnothing\} \tag{9.2-2}$$

where, as defined in Section 2.6.4, A^c is the complement of A and $\varnothing$ is the empty set.

Figure 9.4 shows an example of erosion. The elements of A and B are shown shaded and the background is white. The solid boundary in Fig. 9.4(c) is the limit beyond which further displacements of the origin of B would cause the structuring element to cease being completely contained in A. Thus, the locus of points (locations of the origin of B) within (and including) this boundary, constitutes the erosion of A by B. We show the erosion shaded in Fig. 9.4(c). Keep in mind that that erosion is simply the *set* of



a b c
d e

FIGURE 9.4 (a) Set A. (b) Square structuring element, B. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference.

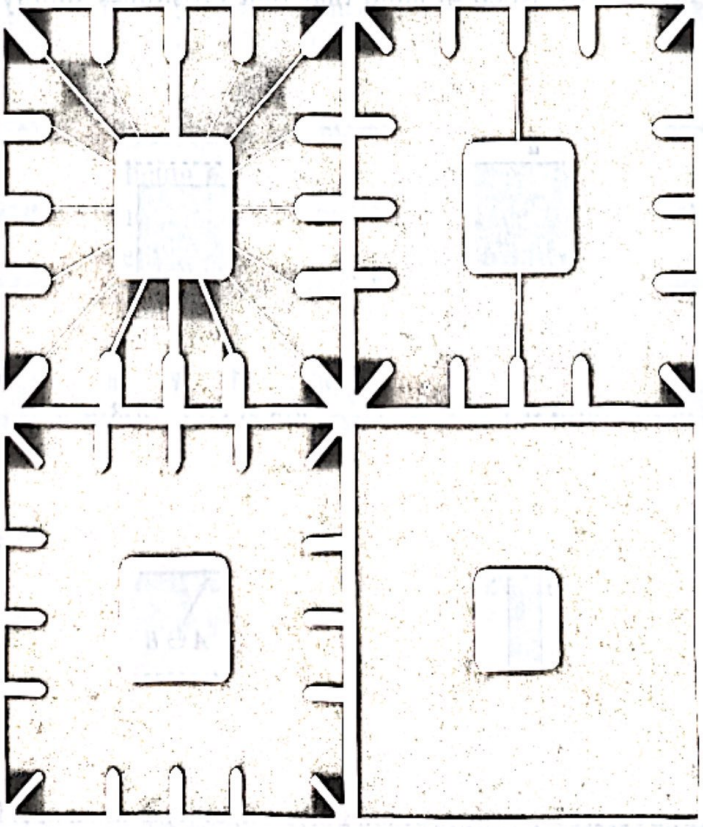
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values of z that satisfy Eq. (9.2-1) or (9.2-2). The boundary of set A is shown dashed in Figs. 9.4(c) and (e) only as a reference; it is not part of the erosion operation. Figure 9.4(d) shows an elongated structuring element, and Fig. 9.4(e) shows the erosion of A by this element. Note that the original set was eroded to a line.

Equations (9.2-1) and (9.2-2) are not the only definitions of erosion (see Problems 9.9 and 9.10 for two additional, equivalent definitions.) However, these equations have the distinct advantage over other formulations in that they are more intuitive when the structuring element B is viewed as a spatial mask (see Section 3.4.1).

EXAMPLE 9.1: Using erosion to remove image components.

■ Suppose that we wish to remove the lines connecting the center region to the border pads in Fig. 9.5(a). Eroding the image with a square structuring element of size 11 × 11 whose components are all 1s removed most of the lines, as Fig. 9.5(b) shows. The reason the two vertical lines in the center were thinned but not removed completely is that their width is greater than 11 pixels. Changing the SE size to 15 × 15 and eroding the original image again did remove all the connecting lines, as Fig. 9.5(c) shows (an alternate approach would have been to erode the image in Fig. 9.5(b) again using the same 11 × 11 SE). Increasing the size of the structuring element even more would eliminate larger components. For example, the border pads can be removed with a structuring element of size 45 × 45, as Fig. 9.5(d) shows.



a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486 × 486 binary image of a wirebond mask. (b)–(d) Image eroded using square structuring elements of sizes 11 × 11, 15 × 15, and 45 × 45, respectively. The elements of the SEs were all 1s.

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We see from this example that erosion shrinks or thins objects in a binary image. In fact, we can view erosion as a *morphological filtering* operation in which image details smaller than the structuring element are filtered (removed) from the image. In Fig. 9.5, erosion performed the function of a "line filter." We return to the concept of a morphological filter in Sections 9.3 and 9.6.3. ■

9.2.2 Dilation

With A and B as sets in Z^2, the *dilation* of A by B, denoted $A \oplus B$, is defined as

$$A \oplus B = \left\{ z | (\hat{B})_z \cap A \neq \varnothing \right\} \tag{9.2-3}$$

This equation is based on reflecting B about its origin, and shifting this reflection by z (see Fig. 9.1). The dilation of A by B then is the set of all displacements, z, such that $\hat{B}$ and A overlap by at least one element. Based on this interpretation, Eq. (9.2-3) can be written equivalently as

$$A \oplus B = \left\{ z | [(\hat{B})_z \cap A] \subseteq A \right\} \tag{9.2-4}$$

As before, we assume that B is a structuring element and A is the set (image objects) to be dilated.

Equations (9.2-3) and (9.2-4) are not the only definitions of dilation currently in use (see Problems 9.11 and 9.12 for two different, yet equivalent, definitions). However, the preceding definitions have a distinct advantage over other formulations in that they are more intuitive when the structuring element B is viewed as a convolution mask. The basic process of flipping (rotating) B about its origin and then successively displacing it so that it slides over set (image) A is analogous to spatial convolution, as introduced in Section 3.4.2. Keep in mind, however, that dilation is based on set operations and therefore is a nonlinear operation, whereas convolution is a linear operation.

Unlike erosion, which is a shrinking or thinning operation, dilation "grows" or "thickens" objects in a binary image. The specific manner and extent of this thickening is controlled by the shape of the structuring element used. Figure 9.6(a) shows the same set used in Fig. 9.4, and Fig. 9.6(b) shows a structuring element (in this case $\hat{B} = B$ because the SE is symmetric about its origin). The dashed line in Fig. 9.6(c) shows the original set for reference, and the solid line shows the limit beyond which any further displacements of the origin of $\hat{B}$ by z would cause the intersection of $\hat{B}$ and A to be empty. Therefore, all points on and inside this boundary constitute the dilation of A by B. Figure 9.6(d) shows a structuring element designed to achieve more dilation vertically than horizontally, and Fig. 9.6(e) shows the dilation achieved with this element.

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a b c
d e

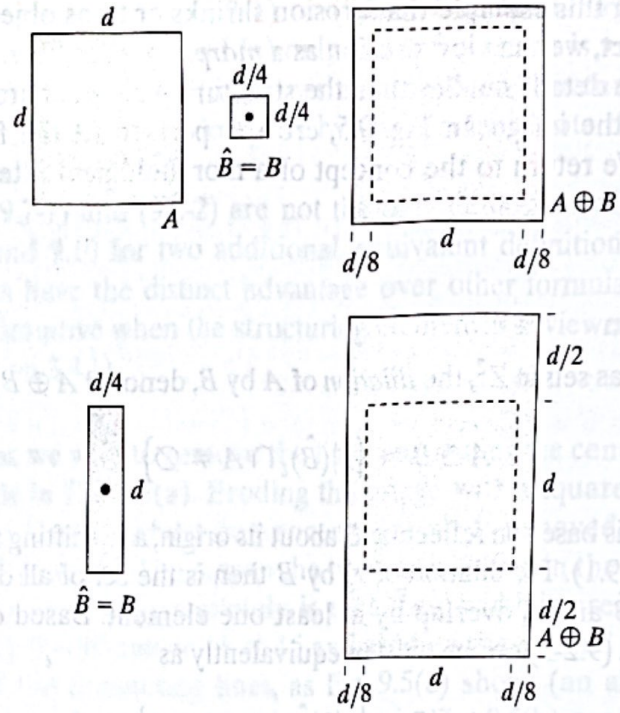
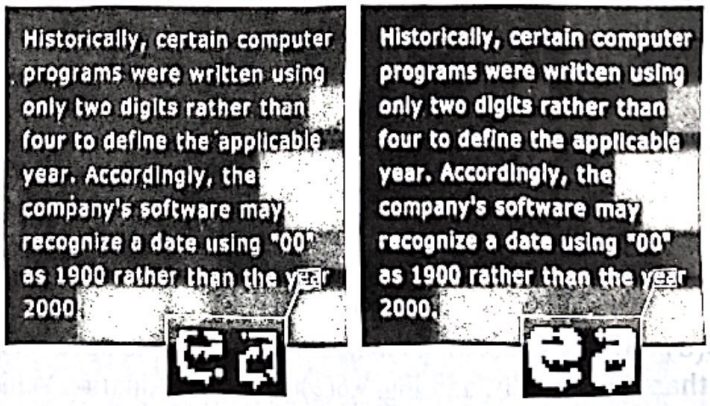


FIGURE 9.6
(a) Set A. (b) Square structuring element (the dot denotes the origin). (c) Dilation of A by B, shown shaded. (d) Elongated structuring element. (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference

EXAMPLE 9.2: An illustration of dilation.

■ One of the simplest applications of dilation is for bridging gaps. Figure 9.7(a) shows the same image with broken characters that we studied in Fig. 4.49 in connection with lowpass filtering. The maximum length of the breaks is known to be two pixels. Figure 9.7(b) shows a structuring element that can be used for repairing the gaps (note that instead of shading, we used 1s to denote the elements of the SE and 0s for the background; this is because the SE is now being treated as a subimage and not as a graphic). Figure 9.7(c) shows the result of dilating the original image with this structuring element. The gaps were bridged. One immediate advantage of the morphological approach over the lowpass filtering method we used to bridge the gaps in Fig. 4.49 is

a c
b



0	1	0
1	1	1
0	1	0

FIGURE 9.7
(a) Sample text of poor resolution with broken characters (see magnified view). (b) Structuring element. (c) Dilation of (a) by (b). Broken segments were joined.

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that the morphological method resulted directly in a binary image. Lowpass filtering, on the other hand, started with a binary image and produced a gray-scale image, which would require a pass with a thresholding function to convert it back to binary form. ■

9.2.3 Duality

Erosion and dilation are duals of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \oplus \hat{B} \tag{9.2-5}$$

and

$$(A \oplus B)^c = A^c \ominus \hat{B} \tag{9.2-6}$$

Equation (9.2-5) indicates that erosion of A by B is the complement of the dilation of A^c by $\hat{B}$, and vice versa. The duality property is useful particularly when the structuring element is symmetric with respect to its origin (as often is the case), so that $\hat{B} = B$. Then, we can obtain the erosion of an image by B simply by dilating its background (i.e., dilating A^c) with the same structuring element and complementing the result. Similar comments apply to Eq. (9.2-6).

We proceed to prove formally the validity of Eq. (9.2-5) in order to illustrate a typical approach for establishing the validity of morphological expressions. Starting with the definition of erosion, it follows that

$$(A \ominus B)^c = \left\{z | (B)_z \subseteq A\right\}^c$$

If set $(B)_z$ is contained in A, then $(B)_z \cap A^c = \varnothing$, in which case the preceding expression becomes

$$(A \ominus B)^c = \left\{z | (B)_z \cap A^c = \varnothing\right\}^c$$

But the *complement* of the set of z's that satisfy $(B)_z \cap A^c = \varnothing$ is the set of z's such that $(B)_z \cap A^c \neq \varnothing$. Therefore,

$$\begin{aligned}(A \ominus B)^c &= \left\{z | (B)_z \cap A^c \neq \varnothing\right\} \\ &= A^c \oplus \hat{B}\end{aligned}$$

where the last step follows from Eq. (9.2-3). This concludes the proof. A similar line of reasoning can be used to prove Eq. (9.2-6) (see Problem 9.13).

9.3 Opening and Closing

As you have seen, dilation expands the components of an image and erosion shrinks them. In this section we discuss two other important morphological operations: opening and closing. *Opening* generally smoothes the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions. *Closing* also tends to smooth sections of contours but, as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour.

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